

Herat Heritage Watch — Semantic Change Classification

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<https://github.com/whamidou006/heritage-watch-release>

Bi-temporal 0.25 m aerial imagery, Herat Old City conservation and buffer zones, Afghanistan (2009–2025)

Herat Old City is on Afghanistan's UNESCO **Tentative** List (ref. 1927, 2004); it is not an inscribed World Heritage Site. Protection is local — municipal conservation and buffer zones administered with the Aga Khan Trust for Culture, with construction inside them requiring Old City Commission permits. Enforcement is weak, which is why an imagery-based record of what was built, demolished or roofed over is worth having at all. **Supporting ablations and the hackathon pack** are in `APPENDIX.md`.

1. Problem description

The data comes as pairs of aerial photographs. Each pair shows the same spot in Herat — one photo taken earlier (t_1), one taken later (t_2) — at a place where an analyst has already marked that something changed. The task is to look at the pair and say which of four changes it was:

- **New Construction** — a building went up
- **Solar Panel** — solar panels appeared, on a roof or on the ground
- **Destruction** — a building came down
- **Temporary Structure** — tents or informal shelter appeared

Scores are macro-F1, which averages the four classes equally so the rare ones still count. Always guessing the most common class scores 0.151, so that is the bar to clear.

These four categories come from the analyst who drew the original annotations, working through the imagery one date-interval at a time from 2009 to 2025.

Destruction and New Construction are the hard part. Both involve the same two scenes — an empty plot and a standing building — and the only thing that separates them is which one came first. Mistake the order and a demolition looks like a new building. This is the central difficulty of the task, not a detail.

Three data properties determine the design:

Property	Consequence
882 usable labels	Too few to train a segmentation network → frozen pretrained encoder + light head
Labels are points, not masks	Dense change-detection heads unusable → classify a patch around each point
0.25 m RGB, 8-bit	Excludes Sentinel/Landsat foundation models (resolution and band mismatch)

A fourth property constrains *evaluation*: labels are **spatially clustered by construction site**, so random cross-validation places neighbouring pixels in both train and test.

2. Dataset

Step 1 — the 21 scenes

All 21 cover the same 2.20×2.19 km tile of Herat on an **identical raster grid** — one distinct bounds/size tuple across all 21 files, 8923×7368 px, EPSG:4326. The tile sits on a degree grid, so pixels are not square on the ground: ≈ 0.247 m/px in x, ≈ 0.297 m/px in y.

Coverage of 2009–2025 is uneven. Twelve years are imaged, **five are missing entirely** (2011, 2014, 2015, 2019, 2020), and some years hold several scenes:

year	scenes	year	scenes
2009	06-12, 06-23	2018	01-04, 02-22
2010	02-25	2021	03-23, 08-06, 08-25
2012	01-07	2022	05-11
2013	09-21, 09-30, 12-04	2023	04-22, 10-11, 10-16
2016	09-27	2024	08-10
2017	07-19, 11-14	2025	01-26

Step 2 — the annotations

An analyst worked through the archive one interval at a time, producing **12 change layers** and **1,370 points**. Each point marks a place where something changed between the two ends of its layer, and carries one category. Layer names give a **month as well as a year** — `change_between_202205_and_202310` means May 2022 to October 2023, not "2022 to 2023". Eleven of the twelve layers are specified to the month; one gives years alone.

Step 3 — from scenes to pairs

A layer names two *moments*; the archive holds scenes on 21 particular days. Each end is matched to the **scene nearest the stated month, within ± 75 days** (the month token read as the 15th). If no scene falls inside that window the layer cannot be imaged honestly and is dropped rather than approximated. Three consequences matter later:

- **Two layers die here.** 2011→2012 and 2012→2013 both need imagery near October 2012, and the nearest scene is 282 days away. That leaves **10 pairs**. Note what does *not* die: 2010→2011 names November 2011, and although no scene falls in that month, 2012-01-07 is only 53 days later and inside the window — so the pair survives with an after-image stamped **2012-01-07**, not 2011.
- **One of the ten pairs is not month-resolved.** The 2024→2025 layer carries no `YYYYMM` tokens and falls back to the year rule. That is **93 of 884 labels, 10.5 %**; the other nine pairs, 791 labels, are month-resolved.
- Only **12 of the 21 scenes** are ever used. The rest lie in un-annotated years, or within days of a scene already chosen as an endpoint.

Step 4 — the 10 pairs

pair	before → after	gap (days)	n	New Constr.	Solar	Destr.	Temp.	links
2009→2010	2009-06-12 → 2010-02-25	258	43	33	0	3	7	
2010→2011	2010-02-25 → 2012-01-07	681	68	48	0	15	5	✓
2013→2016	2013-09-21 → 2016-09-27	1102	142	93	0	46	3	

pair	before → after	gap (days)	n	New Constr.	Solar	Destr.	Temp.	links
2016→2017	2016-09-27 → 2017-11-14	413	48	32	0	15	1	✓
2017→2018	2017-11-14 → 2018-02-22	100	35	26	0	8	1	✓
2018→2021	2018-02-22 → 2021-03-23	1125	105	64	25	16	0	✓
2021→2022	2021-03-23 → 2022-05-11	414	54	34	16	2	2	✓
2022→2023	2022-05-11 → 2023-10-16	523	197	23	136	5	33	✓
2023→2024	2023-10-16 → 2024-08-10	299	99	15	68	15	1	✓
2024→2025	2024-08-10 → 2025-01-26	169	93	14	65	14	0	✓
total			884	382	310	139	53	

(Counts are the 884 surviving steps 1–3 and the class filter; the edge-clip in step 5 removes a further 2 at the 64 px chip size, giving the 882 used throughout. ✓ marks a pair whose before-image is the previous pair's after-image.)

Two properties of this table drive several later results:

- **The gap is not constant** — 100 to 1,125 days, an 11× spread, median 414. A change observed over three years is a different visual problem from one observed over three months, even when the label word is the same. The two widest gaps carry 247 of 884 points (28 %).
- **The class mix moves with time.** There is **not one Solar Panel label before 2018**; all 310 come from the last five pairs. Early pairs are dominated by New Construction (33/43, 48/68, 93/142), late pairs by Solar Panel (136/197, 68/99, 65/93), and 33 of the 53 Temporary Structure labels sit in a single pair. Knowing the date therefore already narrows the answer — this is the source of the date shortcut measured in §A4.

Step 5 — filtering to usable labels

Filtering step	Remaining
Raw annotations	1,370
– no scene within ±75 days of the stated month (110)	1,260
– point outside the imagery tile (350)	910
– classes Other (25) / Reconstruction (1)	884
– patch window clipped at the raster edge (2 at 64 px)	882

(Shown in the order the code applies it. The two middle steps do not commute in their intermediate counts — filtering by footprint first would attribute 367 to the tile test and 93 to missing imagery — but the total, 884, is the same either way. The last step is the only one that depends on the chip size: a 64 px window clips 2 points, a 128 px window clips 5, which is why results quoted at the superseded 128 px size carry $n = 879$ rather than 882.)

The tile test is a lon/lat bounding-box check against the imagery footprint. It removes 350 points (27.8 % of those surviving pairing) simply because the annotator worked across the whole city while the imagery covers one 2.2 km square. For this archive the rectangle is exact rather than an approximation: all 21 scenes share one footprint, and no scene contains blank or nodata pixels inside it (measured blank fraction 0.0000 on scenes from 2009, 2018 and 2025).

Class distribution (final 882). New Construction 382 (43.3 %), Solar Panel 309 (35.0 %), Destruction 138 (15.6 %), Temporary Structure 53 (6.0 %). Skewed 7.2 : 1, so all metrics are macro-averaged; the measured effect of this imbalance is in [APPENDIX.md §A1](#).

Step 5a — why the month and not the year

Reading only the **year** — latest scene of the first year, earliest of the second — mis-selects whenever an interval does not begin in December and end in January, which is most of them. Both rules yield 9 usable pairs from the 11 month-named layers, but not the same 9: **6 of the 8 pairs usable under both receive different endpoint imagery**, the month rule recovers 2010→2011 which the year rule cannot image at all, and it refuses 2012→2013 which the year rule imaged 282 days off target. Coverage — the fraction of the annotated interval actually spanned by the imagery — for the four worst-affected pairs:

layer	annotated	year rule	month rule
2016→2017	2016-09 → 2017-11	69.2 %	97.0 %
2017→2018	2017-11 → 2018-02	54.4 %	100.0 %
2021→2022	2021-03 → 2022-05	60.8 %	97.2 %
2022→2023	2022-05 → 2023-10	66.0 %	100.0 %
each rule's own 9 pairs, mean		81.9 %	98.8 %

Coverage rewards overshoot, so the worst endpoint offset is reported beside it: median **80 → 8 days**, maximum **282 → 53 days**. All 11 layers are archived in `out/window_coverage.json`. The concern is that a patch keeps its change label while the pixels either side of it no longer span the change — a "New Construction" example in which nothing is yet built. That is the mechanism being guarded against; the affected chips have **not** been inspected to confirm it.

One property corroborates the rule independently of any score. Consecutive pairs should meet, the second scene of one interval being the first of the next. Among the nine month-named pairs, junctions that chain exactly rise from **2 of 8 to 7 of 8**; across all ten pairs of the final manifest the chain holds at **8 of 9**. Nothing in the rule optimises for this.

What it is worth, measured. The two manifests were scored against each other paired — identical labels, spatial blocks, folds and classifier, only the pixels differing — over 8 seeds:

population	n	year rule	month rule	Δ	2·SE	seeds won
all paired points	811	0.6783	0.6850	+0.0067	0.0117	5/8
imagery actually moved	516	0.6901	0.6984	+0.0083	0.0072	6/8

Neither difference is unanimous across seeds, and neither reaches the 0.02 minimum detectable effect of §A2, so **the correction is reported as score-neutral** — adopted because the provenance is right, not because it improves the numbers.

Step 6 — what one sample is

two image patches	128 × 128 px cut from the <i>same</i> pixel coordinates in the before- and after-scene (≈32 m across, ≈38 m tall on the ground)
a label	one of the four change classes
a location	longitude / latitude, used only to build spatial folds
a date pair	which of the 10 acquisition pairs it came from

There is no mask and no box — the annotation is a point, and the patch is a fixed window centred on it.

Because the two scenes share a raster grid (step 1), the two patches are **pixel-aligned by construction**. Co-registration, normally the dominant error source in high-resolution change detection, is eliminated rather than modelled.

3. Models

An **encoder** is a neural network that turns an image patch into a fixed list of numbers summarising its content — edges, rooftops, bare ground, vegetation, texture. That list is the patch's **embedding**, and its length is the **dimension** (d). Here the encoders are pretrained by others on large image collections and are **frozen**: their weights are never updated, so they act as fixed measuring instruments rather than as something we train. The **head** is the only part fitted to Herat data — a small classifier that reads the embeddings of the before and after patches and returns one of the four classes.

The head is a multinomial logistic regression ($C=1.0$, balanced class weights). At $n = 882$, $d = 768$ – 8064 , a linear head is the appropriate capacity — gradient boosting was slower and weaker. Of seven configurations evaluated, three are strongest:

A. DINOv2 ViT-B/14 — generic self-supervised features, strong few-shot linear probing. Patches resized to a fixed 224 px input, ImageNet normalisation, represented as $[e_1, e_2, e_2 - e_1]$. *2304-d, 0.715*.

B. Satlas-MI + SI diff (selected) — SatlasPretrain `Aerial_SwinB_MI`, a Swin-v2-B pretrained on 0.5–2 m/px RGB **aerial** imagery (closest available resolution match); chips fed at their native size, 4-level FPN pooled and concatenated. The multi-image encoder max-pools over time and is therefore **exactly order-invariant** — swapping t_1/t_2 changes its output by 0.0. Since Destruction and New Construction differ *only* by temporal order, MI alone cannot separate them; this variant repairs that by re-injecting an explicit directional term $s_2 - s_1$. *3840-d, 0.779*.

C. Merged: DINOv2 + Satlas-SI — both encoders at both dates plus both differences. It scores marginally *above* B (0.787) at more than twice the dimensionality, but the gap is inside the noise floor and so is not resolvable either way; the second encoder buys nothing measurable. *8064-d, 0.787*.

4. Evaluation data and metrics

Evaluation data. The same 882 labelled patches; no separate hold-out is withheld, because at this sample size cross-validation uses the data more efficiently and yields a variance estimate.

Protocol — spatially-blocked CV. Labels cluster tightly at construction sites, so a random split would put neighbouring pixels of the *same* rooftop in both train and test. To prevent that, the 2.2 km tile is cut into a 6×6 grid and each point is tagged with its cell; the cell ID is then passed as the group to `StratifiedGroupKFold(5)`, which keeps every cell **whole** on one side of the split. No test patch can share a cell with a training patch. Random CV is reported **only** to quantify the leakage it introduces.

The grid is equal-area but the labels are not evenly spread, so the folds are uneven:

cells	36 of 36 populated
points per cell	min 6, median 20, max 142
test-fold size	137 – 245 points (5 folds)
cells per test fold	7 – 8

Every fold contains all four classes (Destruction 23–33, Temporary Structure 9–13 per fold), which is what **Stratified** buys; but the 1.8× spread in fold size is a direct consequence of grouping by cell, and it is part of why the noise floor is as wide as it is.

Does the blocking actually work? Measured. Chips are extracted around points, and the points are dense, so the windows overlap: at 128 px, **664 of 879 rows (75.5 %) share pixels with at least one other row**. Cell-grouping is supposed to keep those pairs on the same side of the split. It does — after blocking, only **3.7 % of test rows (mean 32 of 879)** still share pixels with their own training fold, so **blocking removes about 95 % of the chip overlap**. Purging the remainder is worth nothing measurable:

arm	macro-F1	Δ vs keep	2·SE	wins	rows dropped/fold
keep (the protocol)	0.6929	—	—	—	0
purge test rows leaking into their own fold	0.6923	−0.0006	0.0021	4/8	6.4
purge every overlapping row	0.6895	−0.0034	0.0037	6/8	17.8

Both purges land *within noise*, and both go very slightly **down** rather than up — which is what you would expect if the residual overlap were contributing nothing and the purge were merely removing training data. The headline is not inflated by chip overlap.

What is *not* blocked: time. All 10 acquisition pairs appear on both sides of every fold. The splitting controls *where* a point is, never *when* it was observed, so a model may exploit interval-specific appearance and class mix (§2). §A4 quantifies the resulting shortcut.

How much does that flatter the headline? Measured. We ran a leave-one-interval-out control on the corrected month-resolved manifest: train on the other intervals, test on a held-out one, against a same-sized control trained on data that *does* include the test interval. Training-set size and classifier are held fixed, and — unlike the earlier version of this control — **both arms are spatially blocked**, so the contrast isolates acquisition familiarity rather than re-measuring spatial leakage.

Held-out interval	n	seen	unseen	Δ	wins
2009→2010	41	0.3521	0.3556	+0.0034	2/6
2010→2011	68	0.4661	0.4192	−0.0469	5/6
2013→2016	141	0.4703	0.3786	−0.0917	6/6
2016→2017	48	0.5149	0.4190	−0.0959	5/6
2017→2018	35	0.4765	0.5022	+0.0257	3/6
2018→2021	104	0.4297	0.3497	−0.0800	6/6
2021→2022	54	0.5192	0.4315	−0.0876	4/6
2022→2023	197	0.7077	0.5522	−0.1556	6/6
2023→2024	98	0.6046	0.4713	−0.1332	6/6
2024→2025	93	0.7000	0.6427	−0.0573	5/6
mean				−0.0719	8/10 worse

Mean Δ **−0.0719**, sd 0.0559, 2·SE 0.0353. It clears two standard errors and clears the 0.02 minimum effect size, but **8 of 10 is not unanimous**, so under the rule of §4 it is **not resolved**. The two dissenting intervals are the two smallest tested (35 and 41 labels), which is where the estimate is noisiest — but the rule is the rule, and we do not get to drop the inconvenient rows.

Two things worth stating plainly.

The legacy version of this control was roughly twice as large: -0.1755 at 9/10. The difference is that neither of its arms was spatially blocked. So **about half of what looked like a temporal generalisation penalty was ordinary spatial leakage** being counted twice. This is the clearest example in the report of why a control needs the same blocking as the thing it is controlling.

No interval is clean. Under the month rule the scenes **chain** — the after-image of one pair is the before-image of the next — so every "unseen" arm has already seen one of its two endpoint images through a neighbouring pair. A genuinely date-disjoint hold-out cannot be built from this archive at all. The -0.0719 is therefore a **lower bound** on the cost of a new acquisition, and the only way to measure the real thing is a second site.

The practical reading: the headline in §5 is blocked in space but only partially in time, and generalising to a genuinely new acquisition should be expected to cost **at least** 0.07 macro-F1.

A second estimator of the same quantity, and why it is larger. The released benchmark (`evaluate_interval`) measures this differently: it holds out each interval in turn, takes whole spatial cells as test, withholds those cells from training, and then **pools predictions across all ten intervals** before computing one macro-F1 over the fixed four classes. Its matched control is the same function run with the interval labels shuffled — identical capped training pool, identical cell blocking, no date structure. Measured on the selected configuration at 64 px:

Arm	macro-F1
Headline (§5, spatially blocked only)	0.7792
Interval machinery, interval labels shuffled	0.7392
Interval machinery, real intervals	0.5745

So the smaller, doubly-blocked training pool costs 0.040 and **date novelty costs 0.165**.

That is more than twice the -0.0719 above, and the two are not in conflict — they are different estimators. The table above averages a per-interval macro-F1 taken over *whichever classes appear in that interval*; an interval holding three classes is scored on three. Pooling first and scoring over the fixed four removes that compression and is the stricter, more honest measurement, which is why the released benchmark uses it. Neither number is wrong, but **0.165 is the one to quote**, and the -0.0719 should be read as what the older per-interval estimator reports on the same data.

Both remain lower bounds for the same reason: the scenes chain.

Metric — macro-F1 over four classes, so the 53-sample Temporary Structure class carries the same weight as the 380-sample New Construction class. Majority-class floor **0.151**.

Controls. All preprocessing is fitted **within fold**; the GPU solver reproduces the reference sklearn implementation to ≤ 0.009 macro-F1.

Protocol correction (found during review). `StratifiedGroupKFold` is a *greedy* assignment whose `random_state` only breaks ties: on the 64 px set it returned an **identical partition for every seed** (0/841 predictions changed), and at 128 px only 27/838 changed. (*Both measured on the 838-sample set in use at the time; the finding is a property of the splitter, not of the sample count.*) Reseeding the splitter therefore measured almost nothing, and the ± 0.004 it produced was not a noise estimate. We instead **jitter the grid origin** by a random fraction of a block per replicate, which moves every block

boundary and yields genuinely different — still spatially blocked — partitions. **The true single-configuration noise floor is ± 0.016 , about 4× the original figure.** All comparisons below are **paired**: replicate r uses the same grid for every configuration, so a difference is tested per replicate rather than between two independently noisy means.

Decision rule, calibrated against a stress test. Pairing alone is not enough. Because paired deltas are strongly correlated, a "consistent" win can be arbitrarily small. To size that risk we ran 60 trials in which the two "systems" were disjoint random halves of one feature set: a rule of $\text{wins} \geq 7/8$ and $|\Delta| > 2 \cdot \text{SE}$ declared **20 % of those pairs "resolved"**. A win must therefore clear three bars — unanimous across all replicates, $|\Delta| > 2 \cdot \text{SE}$, and $|\Delta| \geq 0.02$ (a minimum effect size):

decision rule	fraction of stress-test pairs called "resolved"
$\text{wins} \geq 7/8, \Delta > 2 \cdot \text{SE}$	20.0 %
$\text{wins} = 8/8, \Delta > 2 \cdot \text{SE}$	6.7 %
$\text{wins} = 8/8, \Delta > 2 \cdot \text{SE}, \Delta \geq 0.02$	1.7 % (1 of 60)

Every claim in §5 is reported under the calibrated rule. **Read these as acceptance fractions in one stress test on one dataset, not as a validated false-positive rate.** Random halves of a feature set are not guaranteed to be equally predictive, so this is not a true null; 1 of 60 is too thin to support a portable rate; and the trials themselves are not archived in `out/`. What the table does establish is the *ranking* of the three rules, which is the reason the third was adopted. Extrapolating it to a per-hackathon count of expected false wins would be exactly the kind of projection this report refuses elsewhere, so we do not.

Guard against partition tuning — and its limits. The grid-jitter seed is fixed and public so that all teams see identical partitions during development; final scoring runs under an **unpublished seed**. Re-scoring the baseline under a secret seed moved it $0.5949 \rightarrow 0.5989$, well inside the floor, which shows the baseline is not partition-sensitive.

Be precise about what this buys. It removes **partition-specific** tuning only. Because every label is released, model selection, feature selection and any supervised training still see the same observations used for scoring, so the final figure retains ordinary model-selection optimism; a changed seed does not undo that. There is also a concrete leak the harness cannot catch: `evaluate()` refits only the classifier handed to it, so a backbone fine-tuned on all 882 labels *before* embedding would contaminate every fold. The participant pack states this as a rule, but an unbiased final claim would need nested evaluation or genuinely unseen labels — ideally a second site.

Why not a blind hold-out? We measured the alternative. Twenty-five spatially disjoint 20 % hold-outs gave macro-F1 **sd 0.060** with a 0.24 spread, and dropped Temporary Structure to a median of **11** test samples. That single-hold-out sd is $\sim 4\times$ the ± 0.016 single-run CV noise floor (and $\sim 10\times$ the SE of the 8-replicate CV mean, ≈ 0.006), and ranking teams that way would largely rank which blocks they drew. With a single site, replicated blocked CV under a secret seed is the better instrument — though, per the paragraph above, neither option removes model-selection bias.

5. Results

All figures below are measured on the **month-resolved dataset** ($n = 882$, §Step 5a) at the **64 px** chip size selected in finding 8. Means over 8 jittered grid replicates. "Paired Δ " is the per-replicate deficit against

the best mean; "wins" counts replicates in which the better model is ahead. Verdicts apply the three-bar rule of §4 — unanimous, $|\Delta| > 2 \cdot \text{SE}$, and $|\Delta| \geq 0.02$.

Representation	dim	Macro-F1	sd	Paired Δ vs C	wins	verdict
Satlas-MI (order-invariant)	1920	0.6881	0.0092	−0.0993	8/8	resolved
DINOv2, post-event only	768	0.6921	0.0113	−0.0953	8/8	resolved
DINOv2, both + diff	2304	0.7147	0.0063	−0.0726	8/8	resolved
Satlas-SI, post-event only	1920	0.7204	0.0050	−0.0669	8/8	resolved
Satlas-SI, both + diff	5760	0.7559	0.0071	−0.0314	8/8	resolved
B · Satlas-MI + SI diff (selected)	3840	0.7788	0.0039	−0.0085	7/8	not resolved
C · Merged DINOv2 + Satlas	8064	0.7873	0.0082	—	—	—

On the choice between the top two. The merged encoder has the higher mean (0.7873 vs 0.7788) and wins 7 of 8 paired replicates, but the gap is **+0.0085** — less than half the 0.02 minimum effect size, not unanimous, and smaller than its own replicate-to-replicate spread (sd 0.0093). The decision rule of §4 returns *within noise*. The two are therefore **not separated**, and we do not claim the merged encoder is better. We select Satlas-MI + SI diff on **parsimony**: equal measured performance at **less than half the dimensionality** (3840 vs 8064) and drawing on a single pretrained *family* (SatlasPretrain Aerial) rather than two unrelated ones. Note the precise claim: the selected configuration still loads **two Satlas checkpoints**, `Aerial_SwinB_SI` and `Aerial_SwinB_MI`, so this is one model family, not one network — we have not measured inference cost, only dimension and dependency count.

(At the superseded 128 px size the same two configurations sat the other way round — B ahead of C by 0.0197, also not resolved. The ordering of the two is not stable across chip size; their inseparability is. That is the substantive point, and it is why neither is claimed as better.)

Pairing is what makes the rest of the table readable: the gaps between adjacent configurations (0.02–0.04) are of the same order as the ± 0.016 single-run noise, so unpaired means could not have separated them.

Leakage from random CV, measured under the original protocol, remains **+2.5 to +6.3 pp**.

Selected model (B), per class. Precision / recall / F1 averaged over folds, with the row-normalised confusion matrix:

Class	P	R	F1	n	share
Solar Panel	0.942	0.890	0.915	309	35.0 %
New Construction	0.813	0.887	0.849	382	43.3 %
Temporary Structure	0.721	0.585	0.646	53	6.0 %
Destruction	0.662	0.623	0.642	138	15.6 %
Macro	0.784	0.746	0.763	882	

(Per-class figures are out-of-fold over the fixed 6×6 grid, averaged across three splitter seeds; the confusion matrix is seed 0. The headline 0.7788 uses the 8 jittered grids instead, so the 0.763 here is a different partition protocol, not merely a different averaging of the same folds. The three-seed spread prints as 0.000 and should not be read as a precision claim — three seeds is too few to estimate it; the eight-grid sd of 0.0039 is the usable one.)

true ↓ / predicted →	Destruction	New Constr.	Solar Panel	Temporary
Destruction	62.3 %	29.0 %	6.5 %	2.2 %

true ↓ / predicted →	Destruction	New Constr.	Solar Panel	Temporary
New Construction	7.9 %	88.7 %	1.6 %	1.8 %
Solar Panel	1.3 %	9.1 %	89.0 %	0.6 %
Temporary Structure	18.9 %	18.9 %	3.8 %	58.5 %

Destruction is now read correctly more than twice as often as it is misread as New Construction — 62.3 % against 29.0 %, a margin of 33.3 points on 138 samples. An earlier version of this report made the *reversed* ordering (43.7 % correct vs 46.8 % misread) its headline failure.

What fixed it was chip size, not the imagery correction and not the choice of encoder. The three candidate explanations can be separated because all four cells of the 2 × 2 have been measured on the same corrected manifest:

	128 px	64 px
C · merged	40.6 % correct / 51.4 % misread — <i>reversed</i>	67.4 % / 27.5 %
B · MI + SI diff	50.0 % / 41.3 %	62.3 % / 29.0 %

Holding the encoder fixed and changing only the chip size is worth **+26.8 points** of correct Destruction recall on the merged encoder; holding the chip size fixed at 128 px and changing only the encoder is worth **+9.4**. Chip size is roughly three times the lever that representation is, and at 64 px *both* encoders get the ordering right. The imagery correction explains none of it: its only clean estimate remains the paired experiment in §Step 5a, **+0.0067, inside the noise floor**.

The confusion remains the largest in the matrix, and the direction of the residual error is unchanged — **both minority classes fail toward the majority class they physically resemble**, though Temporary Structure's leak into New Construction has fallen from 30.2 % to 18.9 %.

Caveat on Solar Panel (§A4). Its 0.915 is partly *temporal*, not purely visual: there are **zero** solar-panel labels before 2018, so the acquisition pair alone is partially diagnostic. On this same four-class population and chip size, an imagery-free control using only the one-hot date reaches macro-F1 **0.4260** against a **0.1511** majority floor — a real and substantial shortcut.

But the model does not appear to be using it. Supplying the date *alongside* the imagery moves the score by **+0.0010** (0.7886 → 0.7896), a twentieth of the noise floor. That is consistent with the imagery already encoding whatever the date would contribute, and it bounds the risk — but it does not prove the Solar Panel F1 is shortcut-free, because a model can exploit a correlate of the date without being handed the date.

Findings.

- 1. Spatial blocking is essential, and it is sufficient.** Leakage inflates every configuration by 2.5–6.3 pp, so random-CV scores on this dataset are not trustworthy. But the blocking that fixes it is doing its job: 75.5 % of chip windows overlap another row's, and cell-grouping removes about **95 %** of that overlap, leaving 3.7 % of test rows touching their own training fold. Purging even those is worth **−0.0006** (4/8) and purging every overlapping row **−0.0034** (6/8) — both within noise, and both slightly negative. The headline is not inflated by chip overlap (§4).
- 2. Pretraining domain does *not* measurably outweigh temporal input — claim withdrawn.** An earlier version of this report stated that Satlas-SI with a single date beats DINOv2 with both dates

plus their difference "in 8/8 paired replicates". That was an error: the 8/8 figure belonged to each configuration's comparison against the *selected* model, not to this head-to-head. Measured directly, the pair is **+0.0189, 6/8** on the old manifest and **+0.0042, 4/8** on the corrected one — inside the noise floor in both cases. Satlas-SI does lead on the mean, but this dataset cannot resolve it. (*What does survive: every Satlas variant outranks its DINOv2 counterpart, and the selected model uses no DINOv2 features at all — see finding 4.*)

3. **Order-invariance is harmful exactly where it matters.** Restoring the directional term $s_2 - s_1$ to the order-invariant Satlas-MI encoder is worth **+0.0908 macro-F1, 8/8 paired, resolved** (0.6881 → 0.7788) — the largest architectural effect measured here. Since Destruction and New Construction differ *only* by temporal order, a symmetric encoder is structurally unable to separate them, and the fix is to re-inject direction explicitly.
4. **A second encoder adds nothing that can be measured.** The merged DINOv2 + Satlas stack (8064-d, 0.7873) is ahead of Satlas-MI + SI diff alone (3840-d, 0.7788) by **+0.0085, 7/8** — below the 0.02 bar and below its own replicate spread, so *within noise*. On the previous manifest the merged stack appeared to lead by a wide margin, and "the encoders are complementary" was reported as a finding; at the superseded 128 px size the ordering reversed again, to 0.0197 *against* the merged model. Across all three measurements the one stable result is that the two cannot be separated. **One well-matched aerial encoder is enough.**
5. **Difference imagery alone is the weakest signal**, below the post-event image alone (§A7). Four-class macro-F1 for f2–f1 is 0.566 on DINOv2 and 0.637 on Satlas, against 0.645 and 0.669 for the post-event image: paired, post-only wins by +0.0788 (8/8, resolved) and +0.0316 (7/8, suggestive). Much of this taxonomy is readable from appearance rather than from change.
6. **Class imbalance redistributes performance but does not cap it** (§A1). Reweighting the loss is worth +0.011 macro-F1 — below the 0.02 floor, so unresolved — and what it actually does is move score between classes: Temporary Structure gains +0.047 F1 and +0.082 recall, paid for by New Construction. Changing the support ratio is starker still: downsampling New Construction from 352 to 52 swings its own F1 from 0.785 to 0.395 and Destruction's from 0.497 to 0.749, while macro-F1 never leaves 0.683–0.710. The binding constraint is directional confusability, not label counts.
7. **Cleaning affects the metric far more than the model** (§A2). The headline contrast — cleaned four-class 0.6961 against raw six-class 0.5767, a gap of **+0.1194** — is mostly a metric artefact: scoring the *same* raw six-class predictions over only the four target classes gives 0.6790, so **+0.1023** of the gap is the choice of which classes the metric averages over. The genuine modelling effects are small: **–0.0172** for keeping the discarded classes in training (scored on the target classes), and **–0.0060** for deduplicating to one label per 32 m. That last number is the useful one — it halves the evaluated population (879 → 442) and removes every near-neighbour, yet costs less than a third of the noise floor.
8. **Chip size matters more than anything else we varied** (§A3) — more than the encoder, and more than the imagery correction. Measured strictly paired — the 871 locations common to all four caches, one shared spatial grid, identical folds, 8 replicates, so the *only* thing that differs is the pixels:

chip	ground	macro-F1	vs 64 px	2·SE	wins	verdict
32 px	8 m	0.7554 ± 0.0058	+0.0272	0.0042	8/8	resolved
64 px	16 m	0.7827 ± 0.0037	—	—	—	best
128 px (<i>used elsewhere</i>)	32 m	0.6942 ± 0.0041	+0.0885	0.0003	8/8	resolved
256 px	64 m	0.6447 ± 0.0028	+0.1380	0.0028	8/8	resolved

64 px beats 128 px by **+0.0885**, about 4.4× the 0.02 floor, with a paired standard deviation of 0.0004 — the gap is near-identical on every partition. This supersedes an earlier unpaired sweep that put the same contrast at +0.019, 7/8, *not resolved*; that sweep compared arms with different populations (832–842 points) and re-normalised the jittered grid to each population's own extent, which is what hid the effect.

The mechanism is **ground extent, not input resolution**. The DINOv2 path interpolates every chip to a fixed 224 px model input regardless of how much ground it covers, so its arms differ only in field of view — and it improves just as much (post-event only 0.6439 → 0.6921; both + diff 0.6656 → 0.7147 going from 128 px to 64 px). A 16 m window is roughly one building and fills the frame; at 32 m the target occupies a quarter of the frame and the neighbours fill the rest. The label is a point, not a footprint, so the wider the window the more of it is somebody else's building. 9. **A modest date shortcut exists, and the model does not use it** (§A4). Class labels correlate with the acquisition pair — Solar Panel is absent before 2018, and 20 of 25 *Other* labels come from a single annotation layer. On the same four-class set as the headline (n = 882, 64 px), the acquisition pair *alone*, with no imagery at all, reaches macro-F1 **0.4260** against a 0.1511 majority floor, so the shortcut is real and substantial. But adding the date as an extra input alongside the imagery moves the score by **+0.0010** — 0.7886 to 0.7896, a twentieth of the noise floor. The shortcut is available; the classifier is not taking it.

1. **Bi-temporal input does not help the damage class** (§A7). Against *microsoft/haste*'s single-date design, the post-event image *alone* separates Destruction from New Construction just as well as the full bi-temporal stack: 0.555 vs 0.535 F1 on Satlas, 0.421 vs 0.430 on DINOv2. All four contrasts (two encoders × F1 and AP) fall **within noise** — the largest is 0.019 against a 2·SE of 0.015, and the signs disagree — so we cannot separate them. Rubble is visible without the date order. What the date order *does* buy is measurable but lands elsewhere. On the four-class task bi-temporal beats post-only by +0.0221 macro-F1 on Satlas (8/8, resolved) and +0.0230 on DINOv2 (7/8, suggestive) — yet on Destruction itself the gain is −0.004 and +0.019, i.e. nothing. The macro gain comes from Temporary Structure (+0.075 on Satlas) and New Construction (+0.043 on DINOv2). The pre-event image alone is *much* worse — bi-temporal beats it by +0.115/+0.112 macro-F1, 8/8 resolved on both encoders — as is the difference image f2–f1 (+0.054/+0.102, both resolved). So the signal is post-event appearance, not change. This is consistent with finding 3: Satlas-MI is *symmetric* in the pair and cannot encode direction, whereas selecting the post-event date is itself directional.

Which numbers sit on which dataset. Everything in §1–§5 and findings 2–10 is now measured on the **corrected, month-resolved** manifest. The ablations in §6 and the appendix were run at the superseded 128 px chip size and carry n = 879 rather than 882; they are reported as *relative* effects, which is how they should be read regardless, and finding 8 is the reason to re-run any of them before quoting an absolute number from them.

Population	n	classes	What uses it
Corrected, month-resolved, 64 px	882	4	§5 tables, per-class table, findings 3, 4, 8, 9
Corrected, month-resolved, 128 px	879	4	findings 5, 6, 7, 10; the §4 interval control; APPENDIX.md
Corrected, unfiltered	905	6	the six-class arm of the cleaning study (§A2, finding 7)

Four claims changed when they were re-measured, which is the main reason the scope of each number is stated at all: findings 2 and 4 changed on the imagery correction, finding 8 reversed outright once the arms were properly paired, and finding 10 stopped reproducing altogether. Two of those four had previously been reported as *resolved* under the same decision rule that now rejects them — the rule catches under-powered claims, not wrongly-controlled ones.

Limitations. Destruction and New Construction remain the dominant confusion: 29.0 % of Destruction is read as New Construction, against 62.3 % read correctly — still the largest error in the matrix, on only 138 samples. The split of F1 between the two (0.642 / 0.849) reflects their 2.8 : 1 support ratio rather than intrinsic difficulty (§A1). **Destruction is the weakest class at 0.642**, with Temporary Structure just above it at 0.646 on only 53 samples, leaking 18.9 % into New Construction. Part of Solar Panel's 0.915 is a **date shortcut** rather than visual skill (§A4). No pretrained head predicts "Solar Panel" — pretraining supplies features, not classes. Encoders are frozen and untuned, so these figures are lower bounds. Finally, the ± 0.016 noise floor means any future claim below ~ 0.03 must be tested **paired**, not by comparing means.

Next steps. Items 2 and 3 of the previous version are done and folded into §4 and finding 9. What remains, in priority order:

1. **Re-run the `APPENDIX.md` ablations at 64 px.** Finding 8 makes chip size the largest single lever measured here, and every appendix ablation predates it. Their *directions* are unlikely to change — most are large — but none of their absolute numbers should be quoted.
2. **Measure on a second site.** The interval control in §4 establishes that a new acquisition costs at least 0.165 macro-F1, but it cannot do better than a lower bound, because the scenes chain and no date-disjoint hold-out exists in this archive. Every remaining question about generalisation is blocked on data, not on method.
3. **Ablate the Solar Panel date shortcut directly** — how much of its 0.915 survives when the acquisition pair is made uninformative. Adding the date as an input is a null (+0.0010), which bounds the risk but does not settle it, since the model can use a correlate of the date without being handed the date.

On method, the ceiling is likely directional discrimination rather than data volume: finding 3 shows the signal is present but under-exploited, and §A1 shows more labels would mainly move F1 between the confusable pair — though note §A1 removes New Construction points from training *and* evaluation, so it does not estimate the value of acquiring more Destruction labels against a fixed benchmark, and a fixed-test-set learning curve is needed to settle that. §A7 suggests the target is a **better encoder of the post-event state** (fine-tuning the aerial backbone on rubble-vs-structure) rather than more temporal channels, since the post-event image alone already outperforms the bi-temporal stack on this pair. That is a promising hypothesis from frozen-feature experiments, not a demonstrated necessity: it does not exclude gains from temporal fine-tuning, a different fusion operator, or better early imagery. Supporting routes remain SAM-based **AnyChange** proposals to decouple localisation from classification and **Changen2 / ChangeStar2** synthetic ordered pairs.

6. Related work

Two 2026 studies from Microsoft AI for Good, Iconem and Planet Labs address heritage monitoring in Afghanistan on the same national corpus of **1,943 archaeological sites** (898 looted, 1,045 preserved) using PlanetScope monthly mosaics at **4.7 m/px**. Neither covers Herat, and neither is a usable baseline here — our imagery is $\sim 19\times$ finer (0.25 m/px aerial) and our label space is four-way semantic typing rather than binary looting or event timing. They are, however, the closest published work, and three of their results bear directly on ours.

	Tadesse et al., <i>Satellite-Based Detection of Looted Archaeological Sites</i> (arXiv 2602.19608)	Tadesse et al., <i>WATCH: Wide-Area Archaeological Site Tracking</i> (arXiv 2605.08160)
task	binary: looted vs preserved	month-level localisation of a change event

	Tadesse et al., <i>Satellite-Based Detection of Looted Archaeological Sites</i> (arXiv 2602.19608)	Tadesse et al., <i>WATCH: Wide-Area Archaeological Site Tracking</i> (arXiv 2605.08160)
data	PlanetScope 4.7 m/px, 2016–2023	PlanetScope 4.7 m/px, 2017–2024
method	CNN on RGB patches vs classical ML on handcrafted features and foundation-model embeddings	three scorers — TED (training-free), SSCD (self-supervised), WS (weakly supervised) — over six foundation models
headline	ImageNet CNN + spatial masking F1 0.926 ; best foundation-model pipeline 0.710	TED+SatMAE 55.0 % exact-month (best at m=0) and 85.0 % within ± 3 months; 92.5 % within ± 3 months is reached by TED+CLIP / GeoRSCLIP / Satlas, not by SatMAE
protocol	site-level stratified split, 5-fold, PCA fit on training folds only	tolerance-based recall, m = 0...6 months

1. Imprecise annotation dates are a known, general problem — not a local defect. WATCH states plainly that heritage ground truth is *"sparse and often temporally imprecise"*, and builds symmetric tolerance margins of up to six months to absorb *"the imprecision inherent in recorded looting dates"*; its recall rises steeply from m = 0 to m = 3. This is the same failure mode as the provenance bug in §Step 5a, and the ± 75 -day tolerance we adopted is a remedy for the same *kind* of problem at a finer scale. The two operations are not equivalent, though — WATCH tolerates error in a *predicted event date*, whereas our tolerance governs *which input image is selected* — so we read this as motivation for taking date uncertainty seriously, not as external validation of ± 75 days.

2. Their encoder ranking is consistent with our selected model. Our selected configuration uses **Satlas features only** — adding DINOv2 buys nothing measurable (finding 4), and every Satlas variant outranks its DINOv2 counterpart on the mean. WATCH ranks **Satlas-Pretrain best or joint-best at relaxed tolerance under all three of its scorers** (92.5 / 92.5 / 90.0), with DINOv3 leading only at exact-month under one scorer, and attributes the gap to geospatially-aligned pretraining outperforming generic visual pretraining. That is the same ordering we observe, on different imagery, resolution and task.

How strongly can we state our side of it? **Matched like-for-like, the family gap is resolved:** Satlas-SI both+diff over DINOv2 both+diff is **+0.0262, 8/8 replicates, 2-SE = 0.0083**, clearing all three bars. What was withdrawn in finding 2 was the narrower and stranger claim that Satlas *single-date* beats DINOv2 *bi-temporal* — an unmatched comparison, which measures **+0.0042, 4/8** and is plainly unresolved. So: the family ranking agrees with WATCH and is measured on matched inputs; the cross-representation shortcut claim is not. The looting paper supplies the corrective in the other direction — there an **ImageNet-pretrained CNN (0.926) beat every foundation-model pipeline (best 0.710)** — so "foundation model" is not by itself a guarantee of anything.

3. Precedent for reporting cross-site transfer qualitatively. WATCH applies its framework to Syria, Turkey, Pakistan and Egypt but explicitly frames this as *"a qualitative operational demonstration rather than a quantitative evaluation"*, because no ground-truth event months exist outside Afghanistan. Any cross-site section for Herat is in the same position and should carry the same caveat.

Where this report is stricter. Neither study reports a measured noise floor, paired replicates, or a minimum detectable effect, and the looting paper's site-level split is stratified but not spatially blocked. The jittered-grid paired protocol with a declared ± 0.016 floor and a null-calibrated decision rule (§4) is a step up on that specific axis — though on tasks different enough that this is a methodological observation, not a performance claim.

Reproducibility: `build_manifest.py --pairing month` → `embed_chips.py / embed_satlas.py` → `compare_jitter.py` (main table), `perclass_month.py` (per-class + confusion), `date_shortcut_month.py` (date control), `month_vs_year.py` (paired imagery-correction test), `loio_control.py` (time-blocking control), `noise_floor.py` (protocol check), `haste_control.py` (HASTE single-date control, §A7), `export_student_manifest.py` (ships hackathon/data/manifest.json). Numeric output for every table in `out/*.json`. Ablations, protocol rationale and the participant pack: `APPENDIX.md`.